Abhishek Bansal

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Github LinkedIn

Education

Indraprastha Institute of Information Technology, Delhi CGPA: 8.32/10
B.Tech. (Computer Science and Engineering)
2022 - present

Vanasthali Public School, Mayur Vihar, Delhi Percentage: 93.8%
CBSE, Standard 12, PCM
2020 - 2021

Vanasthali Public School, Mayur Vihar, Delhi Percentage: 92.6%
CBSE, Standard 10
2018 - 2019

Skills

Expertise Area: ReactJS, NextJS, ExpressJS, NodeJS, HTML, CSS, REST APIs, Android Development, Jetpack Compose, UI/UX Design, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Analysis, Statistical Modeling, Data Visualization, Regression, MySQL, MongoDB, Software Design, Testing, Debugging, Version Control, Object-Oriented Programming, Linux

Programming Languages: Python, Java, C++, C, Kotlin, JavaScript, TypeScript, HTML, CSS

Tools and Technologies: Git/GitHub, MySQL Workbench, Matplotlib libraries, Numpy, Pandas, Scikit-learn, Pytorch, Tensorflow, Yolo, Google Colab, VS Code, IntelliJ, CLion, Ubuntu (WSL), PyCharm, NS-3, Wireshark, Figma, Canva

Technical Electives: Natural Language Processing, Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Algorithm Design and Analysis, Database Management System, Convex Optimization, Artificial Intelligence, Data Science, Computer Vision, Mobile Computing

Work Experience

Full-Stack LLM-Powered Conversational AI (Jan, 25 - May, 25)
Advisor: Dr. Anuj Grover
Technology Stack: Python, LangChain, RAG, ChromaDB, Selenium, NextJS

- Engineered and deployed an LLM-powered conversational agent for the university website using Next.js, FastAPI, and LangChain, integrating ChromaDB for fast retrieval
- Orchestrated a Selenium-driven web crawler with graph-based content modeling to fuel context-rich RAG responses

Freelance Web Developer (Sep, 24 - Feb, 25)
Technology Stack: MERN, NextJs, TypeScript, TailwindCSS

- Crafted responsive, SEO-optimized React & Tailwind sites for 3 clients, introducing interactive product showcases and light/dark modes
- Configured domains and Hostinger hosting, lifting client inquiries by up to 20%
- siskaengineering.com • pragatiindustriesindia.com • madhavahealthpharma.com

Teaching Assistantships, IIIT-Delhi (Aug, 24 - May, 25)

- Conducted tutorials and lab sessions for courses including Introduction to Programming and Data Structures & Algorithms
- Graded assignments and exams, and mentored students under the guidance of Dr. Pankaj Jalote and Dr. Ojaswa Sharma

Projects

Indoor Navigation App and Website
Team Size – 3
(Jan 25 — Apr 25)
GitHub Repository

Technology Stack: Python, OpenCV, YOLO, MiDaS, Kotlin, React

- Engineered a real-time navigation system for visually impaired users, attaining 92% detection accuracy with YOLO and 85% depth estimation using MiDaS
- Optimized Android (Kotlin) and web (React) interfaces to render navigation cues under 150 ms latency

Document Scanning Android App
Team Size – 2
(Jan 25 — Apr 25)
GitHub Repository

Technology Stack: Kotlin, Jetpack Compose, C++, Google ML Kit

- Authored a native Android scanning solution with 98% OCR accuracy and 90% TTS clarity using Google ML Kit
- Automated edge-detection and enhancement, slashing manual crop time by 75%

Automated Music Genre Classification
Team Size – 4
(Aug 24 — Dec 24)
GitHub Repository

Technology Stack: Python, Librosa, Scikit-learn, XGBoost, Random Forest

- Devised a 10-genre classifier on GTZAN with 57 Librosa features, achieving 77% accuracy and 82% F1 after tuning
- Enhanced model by applying dimensionality reduction techniques such as PCA and t-SNE to visualize high-dimensional feature space, preserving 95% variance and enabling better understanding of genre clusters

E-Commerce Website (Myntra Clone)
Team Size – 4
(Feb 24 – May 24)
GitHub Repository

Technology Stack: HTML, CSS, Node.js, MySQL, Python

- Architected a responsive shopping platform with 5 modules (authorization, cart, orders, catalog, payment), boosting mock task success to 88%
- Safeguarded data integrity with transaction controls that eliminated 100% of simulated conflicts

Stick Hero Game
Team Size – 1
(Oct 23 – Nov 23)
GitHub Repository

Technology Stack: Java, JavaFX, FXML, Multithreading, JUnit, OOP, Design Patterns

- Constructed a JavaFX stick-man arcade game with cherry collection at 60 FPS and under 5 ms input latency
- Fortified code quality with JUnit covering 90% of paths and multithreading for 100 concurrent animations

Mini Operating System
Team Size – 2
(Oct 23 – Dec 23)
GitHub Repository

Technology Stack: C, Linux, Git, WSL

- Designed and implemented a priority-based CPU scheduler using C, optimizing process management improving system responsiveness under multi-process workloads
- Streamlined shell and ELF loader with 100% lazy-loading success, cutting memory footprint by 30%

Positions of Responsibility

- **Tech Co-Lead, Techspectra** – Led workshops and hackathons for IIIT-D's tech fest, engaging 10,000+ participants; focused on content design and event execution
- **Coordinator, Foosball Self-Growth Program** – Supervised a training program for 30 participants; managed development, peer-led training, and progress tracking

Awards and Achievements

JEE Advanced: Secured All India Rank 7727 in JEE Advanced. **JEE Mains**: 99.53 percentile – All India Rank 4330 in JEE Mains

Interests and Hobbies

Likes to play chess and foosball

Declaration: The above information is accurate and true to the best of available knowledge.
Abhishek Bansal
Date: July 1, 2025